

Model Paper - Database Management Systems

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Q1 (100 marks)

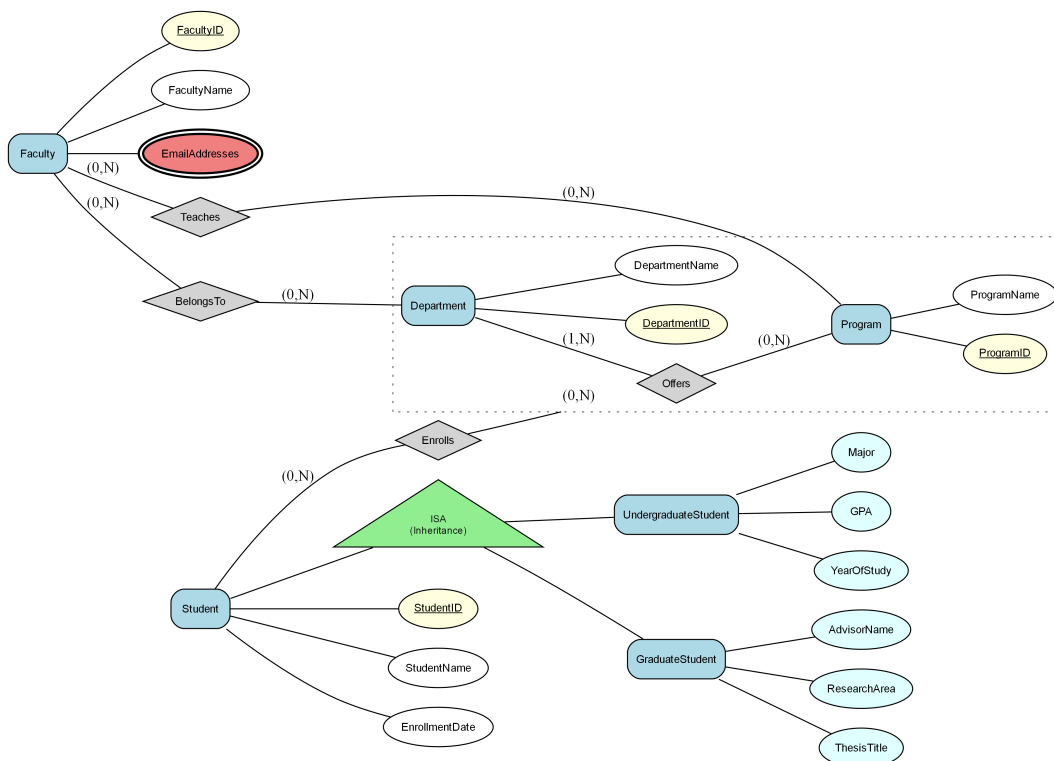
Consider the following Enhanced Entity-Relationship (EER) diagram:

Note: The diagram uses (min, max) cardinality notation where:

- (1,1) indicates one-to-one relationship (each entity participates exactly once)
- (1,N) or (1,*) indicates one-to-many relationship (one entity can relate to many)
- (0,1) indicates optional participation (zero or one)
- (0,N) or (0,*) indicates optional many participation (zero or many)

In a university database system, there are four main entities: Student, Course, Department, and Enrollment. The Student entity has attributes: StudentID, Name (with FirstName, LastName as composite attributes), and ContactNumbers (multivalued attribute). The Course entity has attributes such as CourseID, Title, Credits. contains CourseID, Title, and Credits. The Department entity includes DepartmentID, Name, and Location. The Enrollment entity has attributes such as EnrollmentID, Name, Description. has EnrollmentID, EnrollmentDate, and Status, representing the many-to-many relationship between Student and Course. The Department offers Courses, while Students enroll in these Courses through the Enrollment entity. Furthermore, the Student entity is specialized into two subtypes: GraduateStudent (with additional attributes of ThesisTitle and AdvisorName) and UndergraduateStudent (having YearOfStudy and Major)

Figure: EER



- a) Convert the following EER model into the relational model. Indicate the primary keys and the foreign keys of the resulting relations clearly. (85 marks)
- b) What is the best option for mapping the ISA hierarchies in the diagram? Justify your answer. (15 marks)